



Thermodynamic analysis and prediction of reservoir temperature distribution for steam stimulation

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ABSTRACT

During steam stimulation of horizontal wells in heavy oil reservoir, it is necessary to estimate the reservoir temperature distribution, which has a huge impact on oil production and ultimate oil recovery. Most of previous models were derived based on infinite boundary condition and constant fluid velocity, which were not consistent with real situation of reservoir. In this paper, new energy balance equations combined with mobile boundary conditions were developed for increasing the accuracy of predicted results. Based on dimensionless conversion and numerical analysis, the optimum solution of equation system could be easily obtained, and its good performance in predicting temperature distribution was also validated. Moreover, thermodynamic mechanism of hot fluid, latent heat and sensible heat of steam, and their contributions to heating reservoir were also discussed to investigate the heat transfer pattern in reservoir. Finally, factors that could influence the reservoir temperature distribution were analyzed. It is found that: (a) both the heat conduction and the heat convection contribute to the heat transfer in hot liquid zone, and the heat convection predominates; (b) both the condensation front and heat front move outward with the injection of steam, but their expanding rates gradually drop with time; (c) as the radius increases, the radial fluid velocity will go down, leading the heat convection effect to be weakened and the heat conduction effect to become obvious; (d) there are some factors that can influence the reservoir temperature distribution including steam injection duration, fluid velocity, volumetric heat capacity of fluid, steam quality and thermal conductivity of reservoir.

1. Introduction

With the large consumption of traditional oil and gas resources, the development of heavy oil reservoir has attracted more and more attention (Saripalli et al., 2018). Cyclic steam stimulation (CSS) technology has been widely used as a stimulation method for heavy oil reservoir because of its good performance in enhancing oil recovery. During the CSS process of horizontal wells in heavy oil reservoir, estimating the temperature distribution and the heated area of reservoir plays an important role in predicting crude oil production and assessing the effectiveness of stimulation (Nian and Cheng, 2017; Miah et al., 2018). In this paper, a new model suitable for calculating temperature distribution in the CSS process of horizontal wells was proposed, and this new model is also suitable for steam flooding and steam assisted gravity drainage, as long as the injector is a horizontal well (Baghernezhad et al., 2019).

There are many models used for calculating the reservoir

temperature distribution for the CSS process, the most classic one was created by Marx-Langenheim (Marx and Langenheim, 1959), which calculated the temperature distribution based on energy conservation equation, but Marx-Langenheim's model ignored the impact of steam condensation on the temperature distribution, thereby the result of this model could not illustrate the temperature distribution with high accuracy. After that, many researchers contributed their own wisdom to improve the former model (Willman et al., 1961; Myhill and Stegemeier, 1978). Ramey updated this model by incorporating the variable heat injection rate (Ramey, 1959); Rubinshtein was the first to investigate a method for calculating thermal efficiency (Rubinshtein, 1959); Mandl and Volek improved Marx-Langenheim's model by introducing a critical time of steam condensation (Mandl and Volek, 1969). Prats also made further researches on the basis of Mandl and Volek's model to analyze the impact of heat loss to surrounding formations on reservoir temperature distribution (Prats, 1969, 2001, 2002). Van Lookeren focused on the steam override in formation, and

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presented a new model for calculating the size of steam chamber and oil recovery (VanLookeren, 1983). Based on previous research, Bogerg and Lantz first proposed a new method for predicting oil production and evaluating the thermal effectiveness (Bogerg and Lantz, 1966).

However, previous models mainly consider the expansion of steam chamber, rarely considering steam condensation. In fact, condensed water will pass through the condensation front to form a hot liquid zone in reservoir where the temperature keeps changing with respect to time and distance to wellbore. Hence, Bulter created a new non-isothermal model for steam assisted gravity drainage in which the impact of steam condensation on temperature distribution was taken into account. Whereas this model supposed only heat conduction could make a difference to temperature distribution, ignoring the contribution of heat convection to heat transfer (Butler, 1985). Irani and Cokar modified Bulter's model to increase the accuracy of calculation, but they also neglected the effect of heat convection on temperature distribution (Irani and Cokar, 2016). Li and Cheng initially considered the effect of heat convection on formation temperature distribution, and corrected the steam condensation rate of Sharma and Gates (Li and Chen, 2015). Lauwerier, Spillette, Thomas, LaForce et al. proposed several non-isothermal models for one-dimensional and two-dimensional radial flow, and these models comprehensively considered the effects of heat convection, heat conduction, and heat loss to the surrounding layers on the reservoir temperature, thereby possessing more excellent performance in determining reservoir temperature distribution. However, these models are mainly used in hot water flooding, and the situation of CSS is much more complicated than hot water flooding (Lauwerier, 1955; Spillette, 1965; Thomas et al., 2008; Barends, 2010; LaForce et al., 2014). Yortsos and Gavalas first proposed a Peclet number to characterize the importance of heat convection relative to heat conduction, but this model was developed based on the infinite boundary condition that the temperature of condensed water decreased to initial reservoir temperature at infinity, which was not consistent with the real situation of reservoir and provided us no access to estimate the real heated area of reservoir (Yortsos and Gavalas, 1981).

In this paper, a semi-analytical model was presented to forecast reservoir temperature distribution and heat transfer performance for steam stimulation of horizontal wells. For vertical wells, when steam heats oil layers, thermal energy will lose to overburden and underburden simultaneously (Kaviani, 1995; Ascencio et al., 2014). However, for horizontal wells, at the early stage of steam injection, it takes a long time for steam to reach adjacent layers, leading the energy of the steam to be completely consumed by oil layers, so the heat loss to surrounding layers could be neglected in the new model. In addition, in order to obtain an analytical solution of the partial differential equation, previous models set the fluid velocity in the energy balance equation to be a constant value, which is applicable in one-dimensional linear flow, but when considering the two-dimensional radial flow of steam, the fluid velocity in porous media gradually decreases as radius increases, so previous models are no longer applicable and a new model is necessary to be established.

The new model proposed in this paper has mainly four contributions to the existing body of literature: (a) The heated area of reservoir and the reservoir temperature distribution can be easily estimated; (b) Combined with mobile boundary conditions and new energy balance equations, The prediction accuracy of temperature distribution is largely improved; (c) Thermodynamic mechanism of steam and its heat transfer pattern in reservoir is clearly discussed; (d) Factors that can affect the reservoir temperature distribution during CSS process are analyzed.

2. Methodology

2.1. General assumptions

In order to illustrate the conditions under which this new model could be used, some basic assumptions are made below:

- the reservoir is homogeneous, and fluid flows radially into the oil layer;
- the thickness of the oil layer is large enough, so that the heat loss to surrounding layers can be neglected;
- the rock achieves the fluid temperature instantaneously;
- the thermal energy is fully absorbed by the oil layer and energy conservation law is functional throughout the CSS process;
- the reservoir temperature is constant in steam chamber, decreasing in hot liquid zone, and equal to the initial reservoir temperature in cold oil zone;
- steam heats the formation by means of heat convection and heat conduction;
- the fluctuations of fluid properties and flow parameters along the horizontal section like temperature, pressure and steam quality are neglected

2.2. Description of the new model

The new model can be divided into three zones according to temperature distribution as shown in Fig. 1. The first zone is steam chamber, temperature of which is identical to that of the injected saturated steam. As steam injected, the steam chamber will gradually expand, and the injected steam will condense with steam quality decreasing in this zone. The front of steam chamber is condensation front (CF), at which steam completely condenses into hot water and steam quality decreases to zero. The second zone outside the steam chamber is a hot liquid zone, which is formed by the condensed water. Most of the previous models assume that the condensed water can transfer thermal energy mainly through heat conduction, whereas the new model supposes that both heat conduction and heat convection contribute to the heat transfer in hot liquid zone. In addition, the previous models were developed based on the infinite boundary condition, that is, the temperature of hot liquid will decrease to the initial reservoir temperature at infinity. Therefore, the real size of heated area cannot be obtained

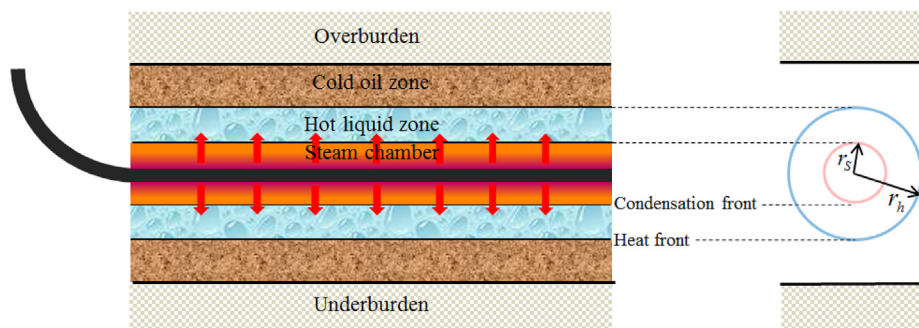


Fig. 1. Illustration of the new model through sectional views; r_s is the radius of steam chamber and r_h is the radius of hot liquid zone.

(Lauwerier, 1955; Spillette, 1965; Thomas et al., 2008; Barends, 2010; LaForce et al., 2014). However, the new model proposes a heat front (HF) of the hot liquid zone and assumes that temperature decreases to the initial reservoir temperature at the HF, so this HF can be used to calculate the real heated area of reservoir. The outermost section of the model is a cold oil zone in which the crude oil has not been affected by the thermal energy, and the temperature of this zone is equal to the initial reservoir temperature.

2.2.1. Size of steam chamber

The thermal energy of steam mainly contains sensible heat and latent heat. As steam diffuses into steam chamber and gradually condenses into hot water, the latent heat of steam will be consumed to heat the relatively cold reservoir. Meanwhile, the latent heat is associated with the steam quality, and the latent heat will decrease during the decrease of steam quality. At the CF of steam chamber, the latent heat has been totally depleted with steam quality decreasing to zero. Based on this assumption, an energy balance equation for the steam chamber is given below,

$$i_s x L_v d\tau = 2\pi l (\rho c)_r (T_s - T_r) r dr \quad (1)$$

Where i_s is the mass injection rate of steam, kg/s; x is the steam quality, decimal; L_v is the latent heat of vaporization, J/kg; τ is the time step, s; $(\rho c)_r$ is the volumetric heat capacity of reservoir, J/(m³·°C); l is the length of horizontal section, m; T_s is the steam temperature, °C; T_r is the initial reservoir temperature, °C; r is the radius, m;

The term on left hand side of Eq. (1) is the total latent heat of steam, and the term on right hand side indicates the thermal energy absorbed by steam chamber. The integral of this equation can be expressed as

$$\int_0^t i_s x L_v d\tau = \int_{r_w}^{r_s} 2\pi l (\rho c)_r (T_s - T_r) r dr \quad (2)$$

Where t is the steam injection duration, s; r_s is the radius of steam chamber, m; r_w is the radius of wellbore, m.

The solution of Eq. (2) can be easily obtained:

$$r_s = \sqrt{\frac{i_s x L_v t}{\pi (\rho c)_r l (T_s - T_r)} + r_w^2} \quad (3)$$

Obviously, the CF of steam chamber will gradually expand with the injection of steam. When the radius of the wellbore is small enough to be neglected, the distance of CF to wellbore (the radius of steam chamber) is proportional to the arithmetic square root of time, which indicates that although the CF moves forward during the injection process, the expanding rate of the CF drops with time.

2.2.2. Temperature distribution of hot liquid zone

In the hot liquid zone, it is the sensible heat of steam that contributes to heat the reservoir. There are two ways for hot liquid to pass through the CF and transfer sensible heat to reservoir, including heat convection and heat conduction (Scott et al., 2005; Hossain and Abu-Khamsin, 2012; Sun et al., 2017). A qualitative representation of reservoir temperature distribution was profiled in Fig. 2. It is obvious that HF is always beyond the corresponding CF, and both the CF and HF are moving ahead with time.

An energy balance equation in cylindrical coordinate system for hot liquid zone is given as:

$$\begin{aligned} \frac{\partial T(\rho c)_r}{\partial t} + \nabla T \cdot (\rho c V)_w &= \nabla \cdot (\lambda_r \nabla T) + q_L \\ t = 0, \quad T &= T_r \\ r = r_s, \quad T &= T_s \\ r = r_h, \quad T &= T_r \end{aligned} \quad (4)$$

Where

$$(\rho c)_r = \varphi [(\rho c)_o S_o + (\rho c)_w S_w] + (1 - \varphi)(\rho c)_{rock} \quad (5)$$

And V_w is the radial water velocity, m/s; λ_r is the heat conductivity of

reservoir, W/(m·°C); q_L is the total heat loss to surrounding layers, J; r_h is the radius of hot liquid zone, m; φ is the porosity of reservoir, decimal; $(\rho c)_o$ is the volumetric heat capacity of oil, J/(m³·°C); $(\rho c)_w$ is the volumetric heat capacity of water, J/(m³·°C); $(\rho c)_{rock}$ is the volumetric heat capacity of rock, J/(m³·°C); S_w is the water saturation, decimal; S_o is the oil saturation, decimal.

Eq. (4) is a partial differential equation (PDE), and the first term on the left hand side (LHS) of the PDE characterizes the cumulative heat in reservoir, while the second term on LHS represents heat convection. For previous models, the fluid velocity 'V' is treated as a constant value in order to bypass the complicity of solving the PDE. In fact, for radial flow, the fluid velocity will gradually decrease with the increase of radius. This paper corrects this term by introducing a new heat convection velocity that changes with respect to radius. Therefore, a new energy conservation equation can be derived. Moreover, the first term on the right hand side (RHS) is heat conduction, and the second term on the RHS denotes heat loss to surrounding layers. Since the heat loss to overburden and underburden is neglected in the new model, the second term on RHS can be set as zero. Hence, Eq. (4) can be simplified as

$$\begin{aligned} \lambda_r \frac{\partial^2 T}{\partial r^2} - \frac{(\rho c)_w Q_w}{2\pi l} \frac{1}{r} \frac{\partial T}{\partial r} &= (\rho c)_r \frac{\partial T}{\partial t} \\ t = 0, \quad T &= T_r \\ r = r_s, \quad T &= T_s \\ r = r_h, \quad T &= T_r \end{aligned} \quad (6)$$

Where Q_w is the volumetric flow rate of condensed water, m³/s.

2.3. Solution of the PDE

By solving the energy conservation equations for the steam chamber and the hot liquid zone respectively, the temperature distribution of each zone can be obtained. However, the energy conservation equation of hot liquid zone is extraordinarily complicated. Therefore, the solving method for the temperature distribution of hot liquid zone is necessary to be discussed in this section.

For CSS, mass flow rate of steam can be converted to volumetric flow rate,

$$Q_s = i_s (1 - x) / \rho_w + i_s x / \rho_g \quad (7)$$

Where Q_s is the volumetric flow rate of fluid (mixture of steam and condensed water), m³/s; ρ_w is the density of water, kg/m³; ρ_g is the density of pure steam, kg/m³.

Since the steam quality reduces to zero in the hot liquid zone, the volumetric flow rate of condensed water can be simplified as:

$$Q_w = i_s / \rho_w \quad (8)$$

Generally, dimensionless conversion can facilitate solving the PDE. Therefore, temperature, radius as well as time can be redefined as:

$$T_D = \frac{T - T_r}{T_s - T_r} \quad r_D = \frac{r}{r_w} \quad t_D = \frac{\lambda_r t}{r_w^2 (\rho c)_r} \quad (9)$$

Moreover, thermal diffusivity is the ratio of heat conductivity to volumetric heat capacity:

$$\alpha = \frac{\lambda_r}{(\rho c)_r} \quad (10)$$

Where α is the thermal diffusivity of reservoir, m²/s.

Therefore, the dimensionless time can be rewritten as

$$t_D = \frac{\alpha t}{r_w^2} \quad (11)$$

Additionally, 'M' is defined as

$$M = \frac{(\rho c)_i i_s}{2\pi l \lambda_r \rho_w} \quad (12)$$

What is worth to emphasize is that 'M' is also a dimensionless

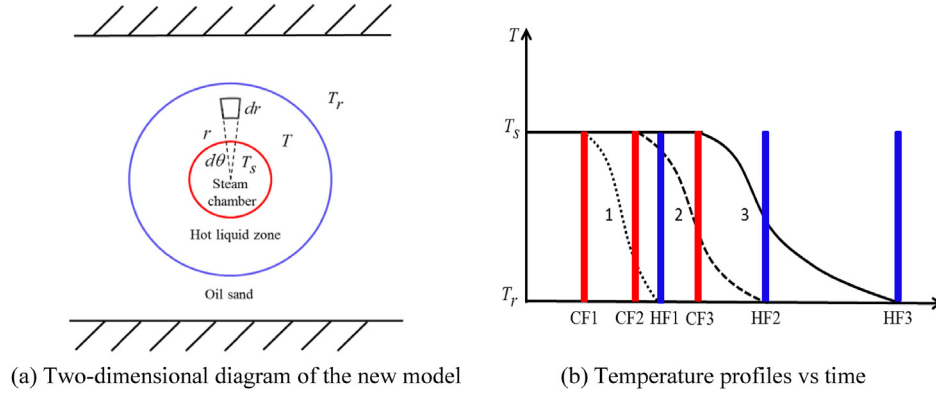


Fig. 2. Illustration of temperature distribution. 1, 2 and 3 demonstrate the different times; CF is the condensation front and HF is the heat front.

parameter, which is the coefficient of heat convection term of the PDE and can be used to characterize the importance of heat convection relative to heat conduction in hot liquid zone. Obviously, the larger the 'M' value, the more the heat convection accounts for total heat transfer.

After dimensionless conversion, the PDE can be rewritten as:

$$\begin{aligned} \frac{\partial^2 T_D}{\partial r_D^2} - M \frac{1}{r_D} \frac{\partial T_D}{\partial r_D} &= \frac{\partial T_D}{\partial t_D} \\ t_D = 0 \quad T_D &= 0 \\ r_{sD} = \frac{r_s}{r_w} \quad T_D &= 1 \\ r_{hD} = \frac{r_h}{r_w} \quad T_D &= 0 \end{aligned} \quad (13)$$

Where r_{sD} is the dimensionless radius of steam chamber; r_{hD} is the dimensionless radius of hot liquid zone.

As mentioned before, the location of HF is still unknown. In order to solve the PDE, another energy balance equation for the hot liquid zone needs to be introduced. The hot liquid zone is heated by the thermal energy of condensed water which is equal to the sensible heat of steam, so the total sensible heat of steam should be equal to the thermal energy absorbed by hot liquid zone, and the absorbed thermal energy can be denoted as the area of the shadow in Fig. 3.

Based on this assumption, another energy balance equation for the hot liquid zone can be expressed as

$$i_s(H_{ws} - H_{wr})t = \int_{r_s}^{r_h} 2\pi r l (\rho c)_r (T - T_r) dr \quad (14)$$

Where H_{ws} is the enthalpy of hot water at the temperature of saturated steam, J/kg; H_{wr} is the enthalpy of hot water at the initial reservoir temperature, J/kg.

The term on left hand side of Eq. (14) is the total thermal energy of condensed water, and the term on right hand side indicates the thermal energy absorbed by the hot liquid zone. After dimensionless conversion,

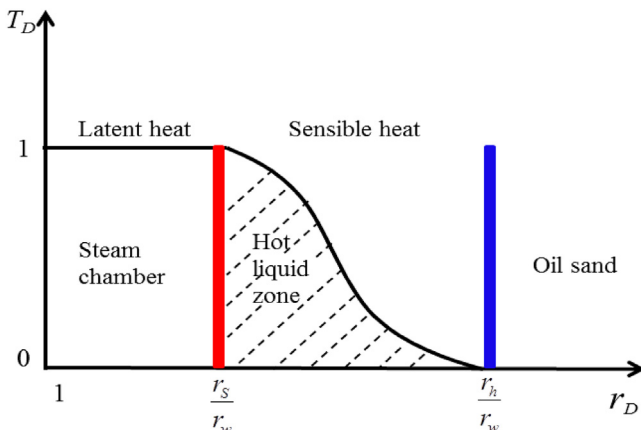


Fig. 3. Dimensionless temperature profile in formation.

the equation above can be expressed as:

$$i_s(H_{ws} - H_{wr}) \frac{t_D (\rho c)_r}{\lambda_r} = 2\pi l (\rho c)_r (T_s - T_r) \int_{r_{hD}}^{r_{sD}} T_D r_D dr_D \quad (15)$$

By solving the set of Eq. (13) and Eq. (15), the temperature distribution of hot liquid zone can be derived.

3. Solving method

Eq. (15) is implicit so that its analytical solution cannot be obtained directly. In addition, Eq. (13) is a second order PDE which is so complicated that its analytical solution is not available either. Consequently, numerical method and trial algorithm method play an important role in deriving the approximate solution. Firstly, if the steam injection duration is given, the CF location can be obtained with Eq. (3) directly, and then a reasonable initial guess of the HF location (the radius of hot liquid zone) can be made in advance. Since both the initial condition and boundary conditions have already been determined, the approximate solution of Eq. (13) can be easily obtained, which can offer us the reservoir temperature distribution at the given condition. After that, the solution of Eq. (13) can be substituted into Eq. (15). If the difference between the value of LHS and the value of RHS is less than the given tolerance, it is considered that an accurate solution is found. Otherwise, another new guess of the HF location ought to be made until an accurate solution that satisfies requirement of tolerance is found. The calculation flow chart is shown in Fig. 4.

4. Result and discussion

4.1. Validation results

In this section, the new method will be compared with Lawal KA's model and simulation results of CMG. Lawal KA's model is one of previous models used for calculating reservoir temperature distribution during steam injection (Lawal and Vesovic, 2010), and the difference between the new model and Lawal KA's model will be exhaustively discussed later.

4.1.1. Basic parameters

The basic parameters used for calculation and numerical simulation are listed in Table 1.

4.1.2. Simulation model

First of all, a three-dimensional simulation model is established in the simulator of CMG as shown in Fig. 5. This model has $70 \times 11 \times 10$ grids, and the step is 5 m in length, 5 m in width and 5 m in thickness in each grid. The horizontal well is perforated from grid (6, 6, 5) to grid (66, 6, 5) in the fifth layer corresponding to the horizontal length of 300 m. In order to analyze the reservoir temperature distribution with

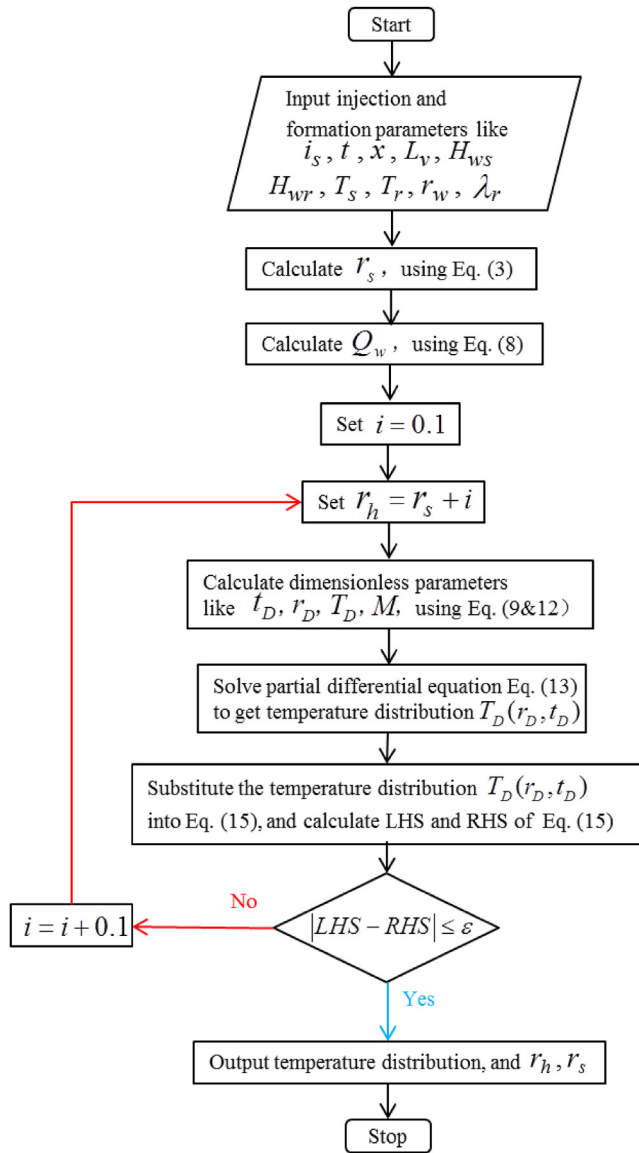


Fig. 4. Algorithm block diagram for solving temperature distribution and heated area.

high accuracy, each grid at near well bore zone has been refined to 400 ($10 \times 10 \times 4$) grids. Therefore, the length and width of these refined grids is 0.5 m, which provides us access to plot accurate temperature profiles based on simulation results. In addition, the grid top of this model is 900 m, and the injection parameters, production parameters as well as reservoir parameters are equivalent to values listed in Table 1.

Table 1

Basic parameters for calculation.

Parameters	Values	Parameters	Values
Initial reservoir temperature (°C)	50	Length of horizontal section (m)	300
Reservoir pressure (Pa)	9.25×10^6	Radius of wellbore (m)	0.1
Absolute permeability (m ²)	1.4×10^{-12}	Steam pressure (Pa)	10.5×10^6
Water saturation (%)	35	Steam temperature (°C)	300
Oil saturation (%)	65	Steam quality (%)	70
Porosity (%)	30	Latent heat of steam (J/kg)	1.4×10^6
Heat capacity of water (J/(m ³ ·°C))	4.2×10^6	Enthalpy of steam (J/kg)	1.34×10^6
Heat capacity of oil (J/(m ³ ·°C))	2.1×10^6	Steam injection rate (t/h)	16
Heat capacity of reservoir (J/(m ³ ·°C))	2.34×10^6	Injection period (d)	15
Heat conductivity of reservoir (W/(m·°C))	2.2	Soaking period (d)	5
Oil density (kg/m ³)	0.95×10^3	Production period (d)	120

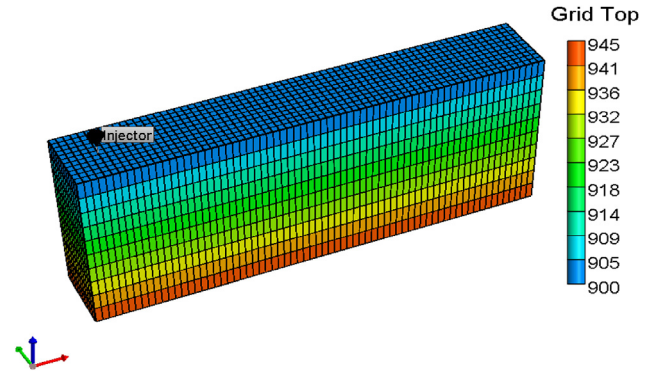


Fig. 5. Three-dimensional simulation model in simulator.

4.1.3. Lawal KA's model

The governing equation and boundary conditions of Lawal KA's model are shown in Eq. (16) and Eq. (17). Although this model takes into account the effect of heat convection on temperature distribution, the fluid velocity involved in convection term ' F ' is a constant value without considering its changing with respect to radius, because this model is founded on the basis of linear flow. In addition, Lawal KA's model uses fixed boundary conditions to solve the PDE, setting the temperature at underburden and overburden as initial reservoir temperature, which is not consistent to the real situation of the formation.

$$\alpha \frac{\partial^2 T}{\partial z^2} + F \frac{\partial T}{\partial z} = \frac{\partial T}{\partial t} \quad (16)$$

$$T(z, 0) = T_i \quad T(0, t) = T_s \quad T(H, t) = T_i \quad (17)$$

Where z is space coordinate, m; F is net convection term, m/s; T_i is the initial reservoir temperature, °C; H is the vertical distance between injection pane and cap rock, m.

4.1.4. Comparison and validation

The temperature distributions obtained from simulation results are plotted in Fig. 6a and Fig. 7a. The predicted results obtained from the new model are compared with simulation results and Lawal KA's model in Figs. 6b and 7b. Apparently, the differences between results from the new model and that from Lawal KA's model can be summarized in two aspects.

Firstly, it is easy to observe from the figures that temperature drops as exponential decline in Lawal KA's model, leading the temperature to decrease more rapidly than the new model. This is because the previous model argues that it is the heat conduction that contributes more to temperature distribution. However, the new model insists that both heat conduction and heat convection can influence the temperature distribution intensively. Consequently, the new model has much smoother temperature profiles. Moreover, the results from the new model also indicate that when the radius is small, the temperature

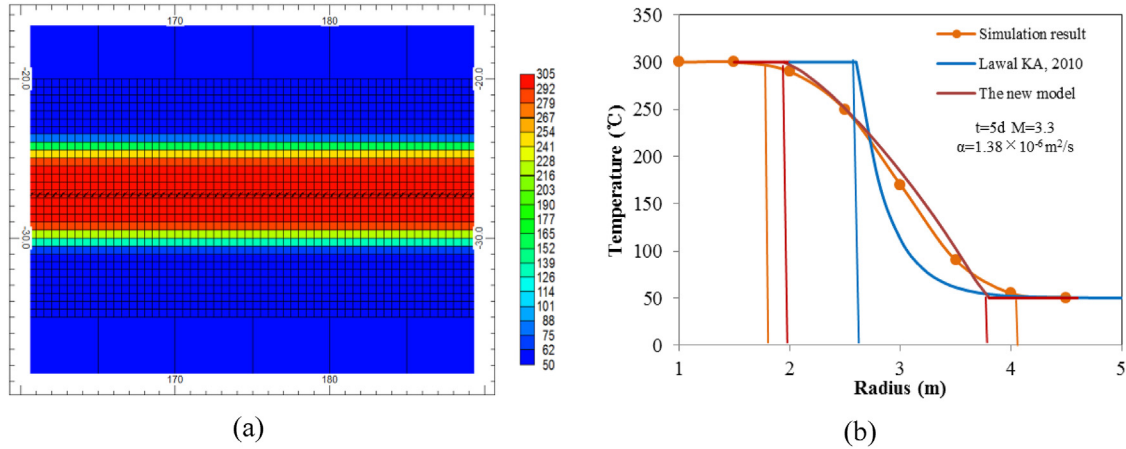


Fig. 6. a) Temperature distribution from simulation results; b) validation of the new model by previous model and simulation results ($M = 3.3$, $\alpha = 1.38 \times 10^{-6} \text{ m}^2/\text{s}$, $t = 5\text{days}$).

decreases slowly, and as the radius becomes larger, the reservoir temperature decreases rapidly. This is because the new model takes into account the change of the fluid velocity. When the radius is small, the fluid velocity is large, so fluid carrying thermal energy heats the formation with high efficiency by means of heat convection. As the radius increases, the fluid velocity drops, causing the heat convection effect to be weakened, and the heat conduction effect to become obvious. Since heat convection is more efficient for heat transfer compared with heat conduction, the ability of fluid to heat the formation is weakened as radius increases.

In addition, Lawal KA's model has fixed boundary conditions, as a result of its assumption that the temperature of hot liquid zone decreases to initial reservoir temperature at cap rock, thereby offering us no access to determine the real heat front of fluid. However, the HF, the mobile boundary of the new model, can predict the heat front directly. Therefore, the amount of heavy oil that has been affected by the thermal energy will be easily determined from the new model. Furthermore, after comparison, the simulation results from commercial numerical simulator indicate that the predicted results of the new model are much more accurate than that of the previous model.

4.2. Characteristic analysis

Dimensionless diagrams that are obtained from the new model and that can demonstrate the reservoir temperature distribution are plotted

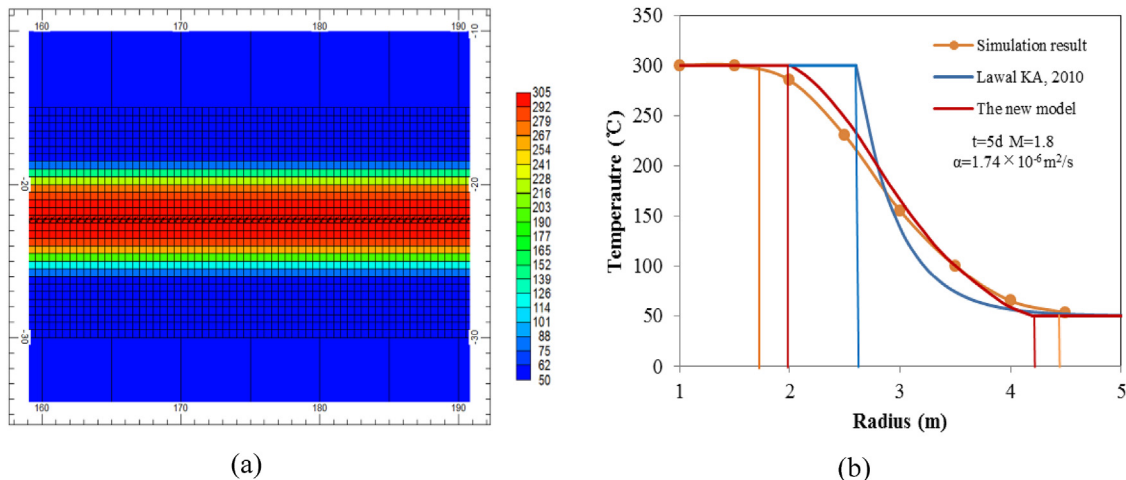


Fig. 7. a) Temperature distribution from simulation results; b) validation of the new model by previous model and simulation results ($M = 1.8$, $\alpha = 1.74 \times 10^{-6} \text{ m}^2/\text{s}$, $t = 5\text{days}$).

in Fig. 8 by MATLAB. The central red surfaces which possess the highest temperature indicate the steam chambers, and the surfaces with varying colors characterize the hot liquid zones.

Obviously, both the steam chamber and hot liquid zone will expand outward with the steam injection, which causes the reservoir temperature to increase; Meanwhile, The reservoir temperature will gradually decrease with the increase of the radius until it reaches the initial reservoir temperature. In addition, compared with Fig. 8a, the predicted results shown in Fig. 8b possess larger areas of hot liquid zone and smoother temperature profiles, as a result of the latter ($\alpha = 1.74 \times 10^{-6} \text{ m}^2/\text{s}$, $M = 1.8$) possessing a larger ' α ' and a smaller ' M ' than the former ($\alpha = 1.38 \times 10^{-6} \text{ m}^2/\text{s}$, $M = 3.3$), when solving the PDE. ' α ' denotes the thermal diffusivity of reservoir, the larger the value, the faster the temperature gradient propagates. Therefore, with the same amount of time, Fig. 8b has larger areas of hot liquid zone; ' M ' characterizes the importance of heat convection relative to heat conduction, and a large ' M ' indicates that the heat convection is prominent in heat transfer. On the contrary, a smaller ' M ' leads the impact of heat conduction on heat transfer to increase. Therefore, influenced by weekend heat convection and intensified heat conduction, Fig. 8b has much smoother temperature profiles.

Fig. 9 and Fig. 10 reflect how the CF and the HF extend over time during the CSS process. Apparently, both the CF and the HF move farther and farther with the steam injection. However, their expanding rate decreases gradually, because for radial flow, the volume of

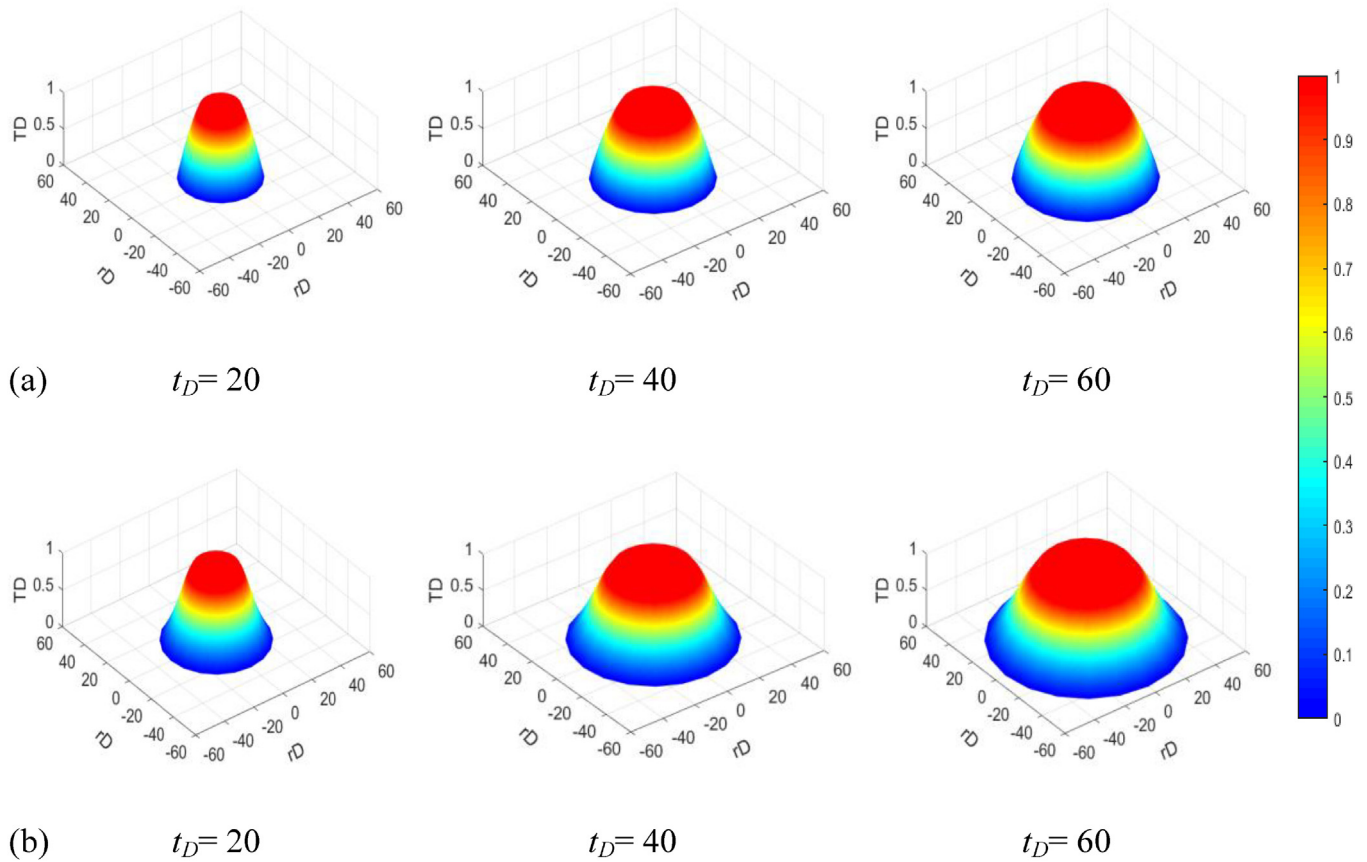


Fig. 8. Dimensionless temperature profiles (T_D) vs dimensionless time (t_D) and dimensionless radius (r_D). a) predicted results while $M = 3.3$, $\alpha = 1.38 \times 10^{-6} \text{ m}^2/\text{s}$; b) predicted results while $M = 1.8$, $\alpha = 1.74 \times 10^{-6} \text{ m}^2/\text{s}$.

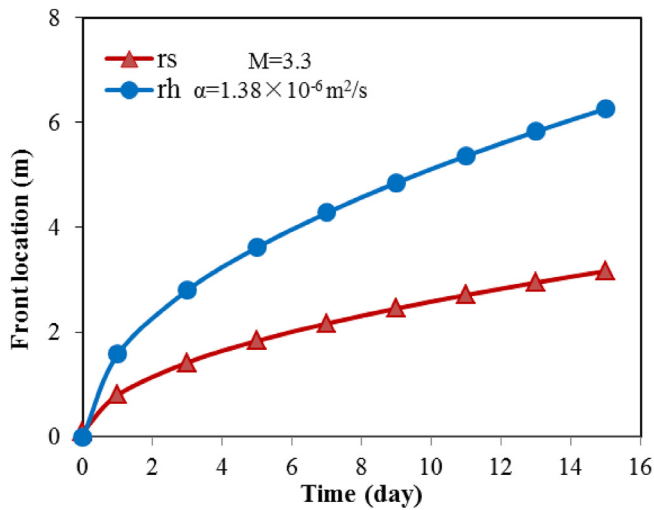


Fig. 9. The radial distance of CF and HF to wellbore vs time in 1st cycle of the CSS process.

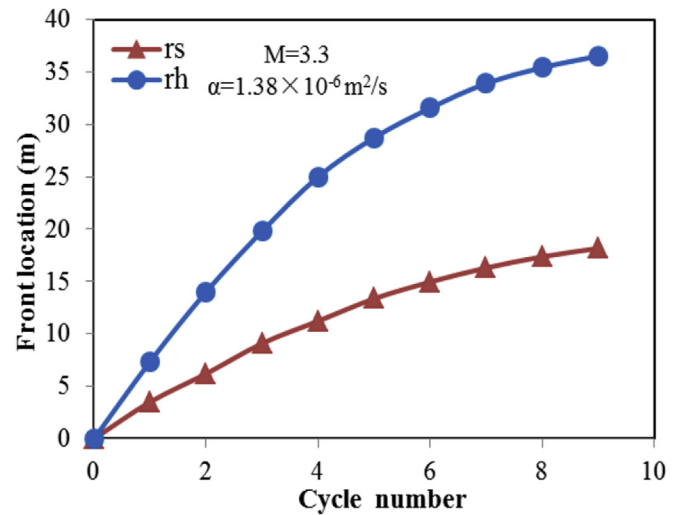


Fig. 10. The radial distance of CF and HF to wellbore vs time at the end of injection period of different cycles.

formation that needs to be swept by hot fluid will rapidly increase with the increase of radial distance, thereby consuming more thermal energy and leading both the CF and the HF to slow down.

4.3. Sensitivity analysis

4.3.1. The analysis of 'M' value

The impact of the 'M' value referred in the PDE on temperature distribution is shown in Fig. 11. It is observed that the temperature

profiles will change from concave-like to convex-like as the 'M' value increases. The 'M' value is related to volumetric heat capacity of fluid, fluid velocity and thermal conductivity of reservoir. A high volumetric heat capacity and a large fluid velocity mean the hot fluid can carry more thermal energy to pass through the CF and heat reservoir by means of heat convection, so the profiles tend to be convex-like. On the contrary, if heat convection weakens and thermal conductivity increases in formation, the effect of heat conduction on temperature distribution will be intensified, and profiles tend to be concave-like.

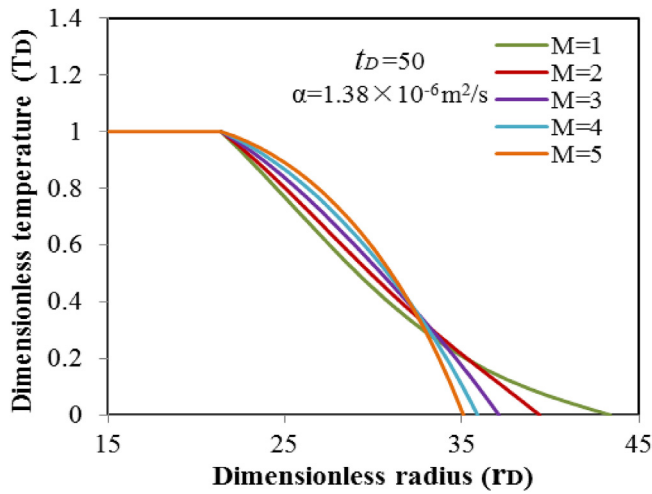


Fig. 11. Effect of 'M' value on temperature distribution.

4.3.2. The analysis of steam injection rate

As mentioned before, the fluid velocity can make a difference to heat transfer. Since the fluid velocity is related to steam injection rate, it is necessary to investigate the effect of steam injection rate on temperature distribution. It is observed from Fig. 12 that the higher steam injection rate, the farther the hot fluid extends, because more thermal energy will be involved in heating the reservoir.

4.3.3. The analysis of heat conductivity

Since different sorts of rocks, like sandstone, mudstone, limestone and shale, have different thermal conductivity, causing diverse heated areas of reservoir, it is necessary to analyze how the thermal conductivity of reservoir influences temperature distribution. From Fig. 13, it is easy to figure out that the reservoir with high thermal conductivity allows hot fluid to diffuse farther, thereby possessing smoother temperature profiles. Nevertheless, when the thermal conductivity doubles, the temperature profiles only stretch slightly, which indicates that the heat conduction accounts a small fraction for heat transfer and the effect of heat conductivity on temperature distribution is not obvious.

4.3.4. The analysis of steam quality

Obviously, the steam quality can make a difference to temperature distribution. It is observed from Fig. 14 that the steam with higher quality contains more thermal energy, causing both the CF and the HF to move farther in formation. What is worth to stress is that the steam

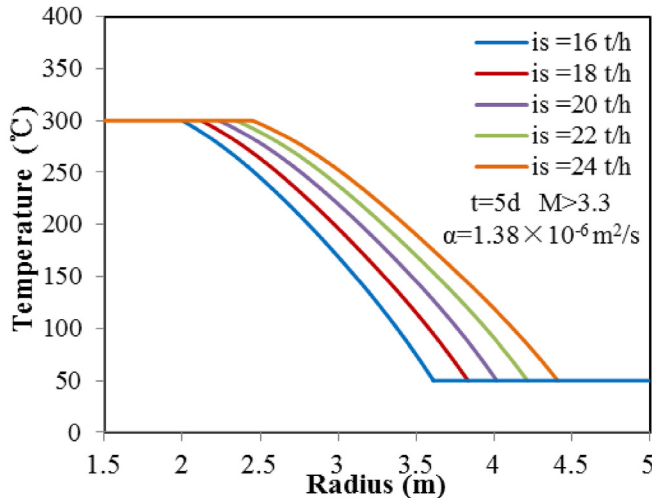


Fig. 12. Effect of steam injection rate on temperature distribution.

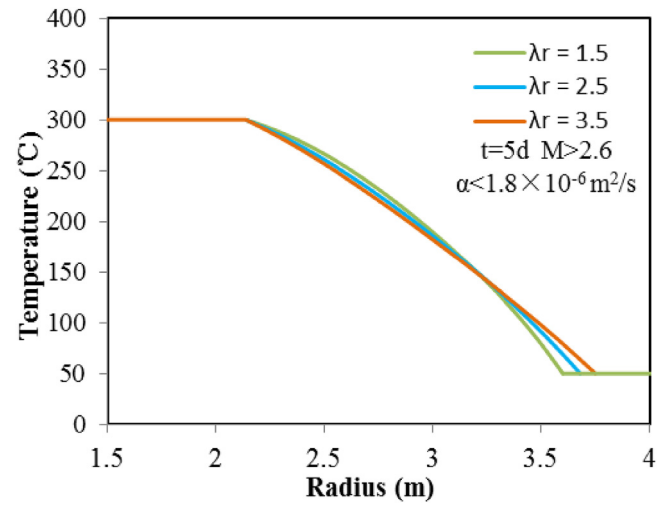


Fig. 13. Effect of heat conductivity of reservoir on temperature distribution.

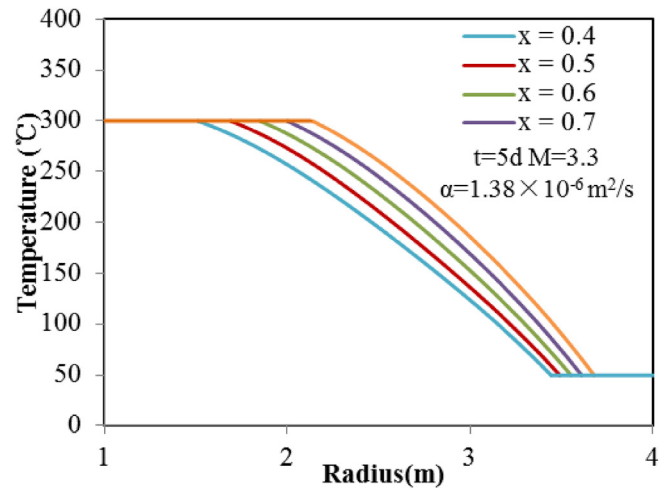


Fig. 14. Effect of steam quality on temperature distribution.

with high quality tends to form a large steam chamber and a relatively small hot liquid zone. On the contrary, the steam with low quality is prone to construct a small steam chamber and a relatively large hot liquid zone. This is because the higher the steam quality, the larger the ratio of latent heat to sensible heat, which affects the relative size of steam chamber and hot liquid zone.

5. Conclusion

In this paper, a novel approach for forecasting reservoir temperature distribution and heat transfer pattern during steam stimulation was presented and validated, and some meaningful conclusions are summarized below:

- With the help of new energy balance equations and mobile boundary conditions, the accuracy of predicted results was substantially improved and the heated area of reservoir could be easily calculated by the new model.
- Condensation front (CF) as well as heat front (HF) will move outward with the injection of steam, but their expanding rates gradually drop with time because of the rapid depletion of thermal energy by reservoir.
- Both heat convection and heat conduction contribute to the heat transfer in hot liquid zone, and as the radius increases, the radial fluid velocity drops, causing the heat convection effect to be

weakened and the heat conduction effect to become obvious, but the heat convection still plays a prominent part in total heat transfer, and the heat conduction only accounts a small fraction for heat transfer.

- (d) There are several factors that can affect the reservoir temperature distribution: the total heated area of reservoir extends further with the increase of steam injection duration; fluid with a large velocity and a high volumetric heat capacity can carry more thermal energy to heat the reservoir, leading temperature profiles to be convex-like and stretch further; the total heated area will grow up as the steam quality increases, and the steam with high quality tends to form a large steam chamber and a relatively small hot liquid zone; last but not the least, the reservoir with high thermal conductivity allows thermal energy to diffuse farther, thereby possessing smoother temperature profiles.

Notes

The authors declare no competing financial interest.

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